BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary Addition and Shifts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Convert 00101101 binary to denary.

2. What is a character set?

A) Characters linked to codes
B) A list of fonts
C) A list of filenames
D) A collection of pixel colours

3. "A record must contain only one unnamed value." Correct this. Explain why.

4. Valid ages are 13-18. Changing <18 to <=18 fixes a bug. Give three tests near 18 and their expected results. Why retest a value that worked before?

Today's objectives

Recall key facts about binary addition and shifts. Work through a new example with correct technical language.

Key vocabulary - Read, cover, recall

Carry
Value passed to the next binary column during addition.

Overflow
Result exceeds the largest value representable in fixed-width storage.

Logical left shift
Move bits left and insert zeros on right.

Logical right shift
Move bits right and insert zeros on left.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Binary addition rules	0+0=0; 0+1=1; 1+1=10, so write 0 and carry 1 into the next column.
Work right to left	Line up bits, add each column including any carry, then record the output.
Logical shifts	A logical left or right shift moves each bit by the stated number of places, introducing zeros at the vacated end.
Overflow and loss	With fixed eight-bit storage, a carry beyond bit 7 is lost (overflow); bits shifted out are discarded.

I Do - One worked example

Add the eight-bit values 00010110 and 00001101, showing the carry where needed. [4 marks]

1. Line up eight positions

2. Add from the right: 0+1=1; next 1+0=1.

3. At the third bit, 1+1=0 carry 1; continue including the carry.

4. The eight-bit result is 00100011 (35 denary). [4]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Calculate 00101101 + 00000111 in binary. [4]

2. Perform a logical left shift of 00010101 by one place. [3]

3. Perform a logical right shift of 10110000 by two places. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the multiplier to model a three-place logical left shift without overflow.

```
value = 9
result = value * 2
print(result)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Add 00011001 and 00000111 as eight-bit unsigned binary. [3]

Q3. Apply a logical left shift of two places to 00001011. [2]

Q4. Explain why adding 11111111 to 00000001 causes overflow in an eight-bit register. [3]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An algorithm only gives reliable results when its inputs are handled as specified. ☐ T ☐ F

2. 0+0=0; 0+1=1; 1+1=10, so write 0 and carry 1 into the next column. ☐ T ☐ F

3. Line up bits, add each column including any carry, then record the output. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 45, because 32 + 8 + 4 + 1 = 45. ☐

Q2 A - Characters linked to codes ☐

Q3 Named fields about one item or person. Fields may have different data types. ☐

Q4 17 accepted; 18 accepted; 19 rejected. Retest a value that worked before to check for regression: a new fault caused by the fix. ☐

Code First - Model

```
value = 9
result = value * 8
print(result)
```

Output: 72

We Do - Check the model

1. 00101101 (45) + 00000111 (7) = 00110100 (52) [4; credit aligned working/carries]. ☐

2. 00101010 [2]; all bits move one position left and 0 enters at the right [1]. ☐

3. 00101100 [2]; two zeros enter on the left [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 00011001 (25) + 00000111 (7) = 00100000 (32) [3]. ☐

Q3 00101100 [2]. ☐

Q4 The mathematical result is 256 [1], requiring a ninth bit [1]; the carry beyond eight bits cannot be stored [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: